



DR ESLAM AHMED
eliteigcse.com

Solving Inequalities

An inequality is an equation with one extra rule: multiply or divide by a negative — flip the sign.
Quadratic? Sketch the curve and read the answer.

THE TOPIC ON ONE MAP

Solving Inequalities

01 — Number lines and integers

$$-2 \leq n < 3: \text{list } n$$

read the diagram

$$\text{draw } y \leq 1$$

02 — Solve a linear inequality

$$7t - 8 < 2t + 7$$

$$3 - 4x \leq 11 \text{ (flip!)}$$

$$4 > 11 - \frac{x}{3}$$

is Martha correct?

03 — Double inequalities

$$-7 \leq 2x - 3 < 5$$

$$-4 \leq 2y < 6, \text{ integers}$$

two conditions at once

04 — Quadratic inequalities

$$6x^2 - 7x - 20 > 0$$

$$4x^2 + 4x - 15 < 0$$

read it from a graph

05 — Problems with inequalities

$$\text{area} < 75 \text{ cm}^2$$

linear + quadratic, list n

greatest value of $y - x$

IN THE EXAM Paper 1 starts with an easy number line; Paper 2 ends with a quadratic inequality for 3–4 marks.

01 Number Lines and Integers

Open circle: the number is **not** included. Filled circle: it **is**. Integers: list every whole number in the gap.

WHEN YOU SEE → YOU DO

$<$ or $>$

→ open circle •

$\leq$ or $\geq$

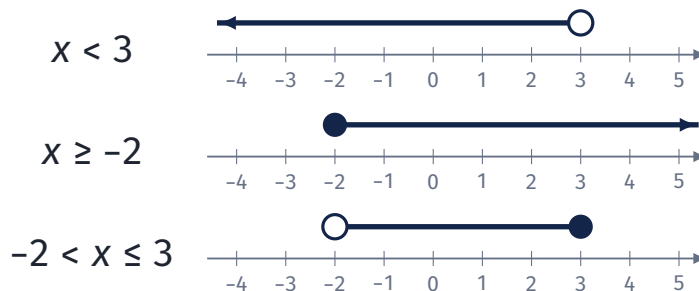
→ filled circle •

“ n is an integer ... write down all the values”

→ whole numbers only — check both ends

“write down the inequality shown”

→ read both circles: $-3 < x \leq 4$



WORKED MODEL • INTEGERS

$-2 \leq n < 3$, n is an integer. Write down all the values of n .

1 -2 is in ($\leq$); 3 is out ($<$)

check the ends

ANSWER $-2, -1, 0, 1, 2$

× **TRAP** Forgetting 0, or including the end with the $<$ sign. Say every integer out loud.

IDEA 01

Practise: Number Lines and Integers

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** List the integers

CH 2 • TOPIC 11 2 marks

$-3 \leq 2n + 1 < 7$, n is an integer. Write down all the values of n .

SELECTED PRACTICE

Q2**PRACTICE**

Practice • Easy

CH 2 • TOPIC 11

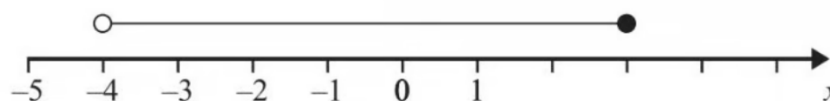
5 marks

2 (a) n is an integer.

$$-1 \leq n < 4$$

List the possible values of n .

(2 marks)

(b)

Write down the inequality shown in the diagram.

(2 marks)

(c) Solve $y - 2 > 5$

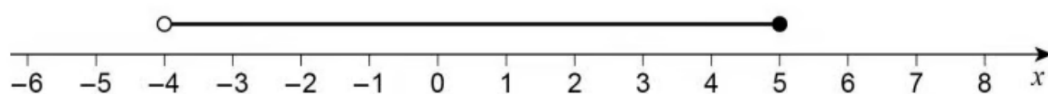
(1 mark)

Q3**PRACTICE**

Practice · Easy

CH 2 · TOPIC 11 1 mark

13 Circle the inequality shown by the diagram.



$-4 \leq x < 5$	$-4 \leq x \leq 5$	$-4 < x < 5$	$-4 < x \leq 5$
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(1 mark)

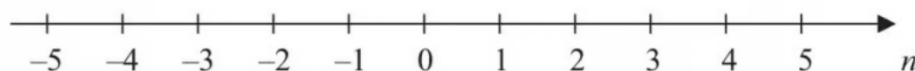
Q4**PRACTICE**

Practice · Medium

CH 2 · TOPIC 11 4 marks

1 (a) $-2 < n \leq 3$

Represent this inequality on the number line.



(2 marks)

(b) Solve the inequality $8x - 3 \geq 6x + 4$

(2 marks)

Q5**PRACTICE**

Practice · Medium

CH 2 · TOPIC 11 5 marks

16 (a) Solve $5x + 6 \geq 3x + 15$

(3 marks)

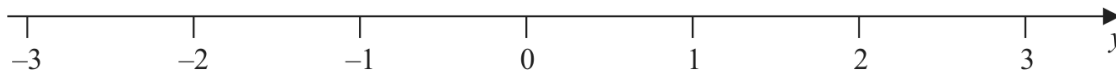
(b) Write down the inequality represented by the number line.

(2 marks)

PAST PAPERS

Q6 **PAST PAPER** Edexcel 4MA1 · January 2021 · Paper 1HR · Q3**CH 2 · TOPIC 11** **4 marks**

- 3 (a) On the number line, show the inequality $-2 \leq y < 1$



n is an integer.

- (b) Write down all the values of n that satisfy $-3.4 < n \leq 2$

.....
(Total for Question 3 is 4 marks)

Q7**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1HR • Q1

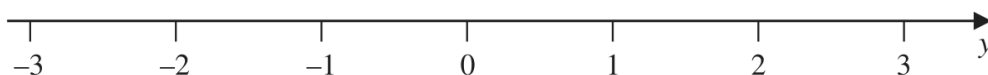
CH 2 • TOPIC 11 3 marks

You must write down all the stages in your working.

1 n is an integer.

(a) Write down all the values of n such that $-2 \leq n < 3$

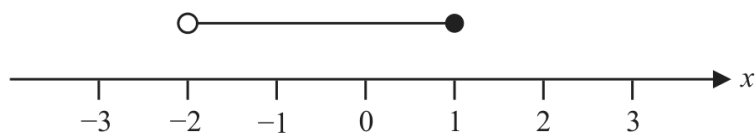
(b) On the number line, represent the inequality $y \leq 1$



(Total for Question 1 is 3 marks)

Q8 **PAST PAPER** Edexcel 4MA1 • June 2025 • Paper 1HR • Q6**CH 2 • TOPIC 11** **4 marks**

6



(a) Write down the inequality shown on the number line.

.....

(b) Solve the inequality $7a - 5 \leq 3a + 28$
Show clear algebraic working.

.....

(Total for Question 6 is 4 marks)

02 Solve a Linear Inequality

Solve it exactly like an equation — Clean, Collect, Divide. One extra rule: $\times$ or $\div$ by a negative flips the sign.

WHEN YOU SEE → YOU DO

“Solve the inequality”

→ same steps as an equation, keep the sign

dividing by -4

→ flip: $\leq$ becomes $\geq$

“largest integer that satisfies”

→ solve, then take the integer below

“Is her solution correct?”

→ solve it yourself, then compare circle and direction

Clean

Collect

Divide

Negative? Flip

WORKED MODEL • JUNE 2022 P1HR

Solve $3 - 4x \leq 11$.

1 $-4x \leq 8$

collect

2 $x \geq \frac{8}{-4}$

$\div$ negative: flip

ANSWER $x \geq -2$

AVOID THE FLIP Move the x term to the side where it stays positive: $3 - 4x \leq 11 \Rightarrow 3 - 11 \leq 4x \Rightarrow -2 \leq x$.

IDEA 02

Practise: Solve a Linear Inequality

My questions first, then selected practice, then real past papers.

Q9**DR ESLAM**

Keep the sign

CH 2 • TOPIC 11

2 marks

Solve $3x + 2 < 17$

Q10**DR ESLAM**

Flip the sign

CH 2 • TOPIC 11

2 marks

Solve $4 - 3x \geq 13$

SELECTED PRACTICE

Q11

PRACTICE

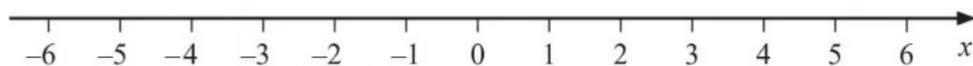
Practice • Easy

CH 2 • TOPIC 11

5 marks

1 (a) Solve $14n > 11n + 6$

(2 marks)

(b) On the number line below, show the set of values of x for which $-2 < x + 3 \leq 4$ 

(3 marks)

Q12

PRACTICE

Practice • Easy

CH 2 • TOPIC 11

2 marks

15 Solve the inequality.

$$3x - 2 > 10$$

(2 marks)

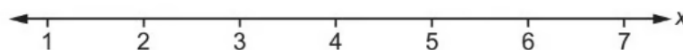
Q13**PRACTICE**

Practice · Medium

CH 2 · TOPIC 11 4 marks

18 Solve $3x + 4 < 19$.

Show your solution on the number line.

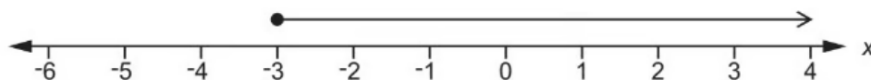


(4 marks)

Q14**PRACTICE**

Practice · Hard

CH 2 · TOPIC 11 4 marks

12 Martha's solution to the inequality $8x + 5 \leq 3x - 10$ is shown on the number line.

Is her solution correct?

Explain your reasoning.

(4 marks)

PAST PAPERS

Q15 PAST PAPER Edexcel 4MA1 • January 2021 • Paper 2H • Q5

CH 2 • TOPIC 11 7 marks

- 5 (a) Expand and simplify
- $4x(2x + 5) - 3x(2x - 3)$

(2)

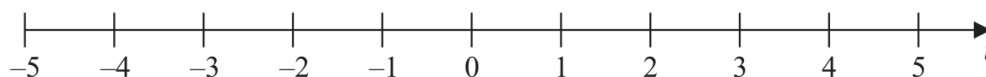
Given that $\frac{y^5 \times y^n}{y^6} = y^{13}$

- (b) work out the value of
- n
- .

 $n =$

- (c) (i) Solve the inequality
- $7t - 8 < 2t + 7$

- (ii) On the number line below, represent the solution set of the inequality solved in part (c)(i)



(Total for Question 5 is 7 marks)

Q16**PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 1HR · Q9

CH 2 · TOPIC 11 2 marks

9 Solve the inequality $3 - 4x \leq 11$

(Total for Question 9 is 2 marks)

03 Double Inequalities

Three parts: left $\leq$ middle $<$ right. Whatever you do, do it to **all three parts** until x is alone in the middle.

WHEN YOU SEE → YOU DO

$$-7 \leq 2x - 3 < 5$$

→ +3 to all three, then $\div 2$ all three

“ y is an integer”

→ solve, then list

two separate inequalities

→ solve each, draw both, keep the **over-lap**

WORKED MODEL • NOVEMBER 2021 P1H

Solve $-7 \leq 2x - 3 < 5$ and show it on a number line.

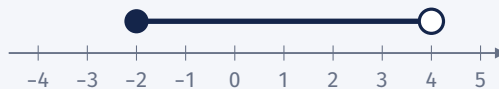
1 $-4 \leq 2x < 8$

+3 to all three

2 $-2 \leq x < 4$

$\div 2$ to all three

ANSWER $-2 \leq x < 4$



IDEA 03

Practise: Double Inequalities

My questions first, then selected practice, then real past papers.

Q17 **DR ESLAM** All three parts

CH 2 · TOPIC 11 3 marks

Solve $-5 < 3x + 1 \leq 10$

SELECTED PRACTICE**Q18** **PRACTICE** Practice · Medium

CH 2 · TOPIC 11 4 marks

4 (a) Solve $6x + 4 > x + 17$

(2 marks)

(b) n is an integer with $-5 < 2n \leq 6$

Write down all the values of n

(2 marks)

Q19**PRACTICE**

Practice • Medium

CH 2 • TOPIC 11 **3 marks**

12 x is an integer.

$$-4 < x \leq 2$$

and

$$2 \leq x + 3 < 9$$

Work out all the possible values of x .

(3 marks)

Q20**PRACTICE**

Practice • Hard

CH 2 • TOPIC 11 3 marks

11 Here are two inequalities.

$$-2 \leq x \leq 3$$

$$9 \leq x + y \leq 11$$

x and y are integers.

Work out the **greatest** possible value of $y - x$

(3 marks)

PAST PAPERS

Q21 PAST PAPER Edexcel 4MA1 · May 2021 · Paper 1H · Q7

CH 2 · TOPIC 11 4 marks

7 $-4 \leq 2y < 6$

 y is an integer.(a) Write down all the possible values of y .

.....

(b) Solve the inequality $7t - 3 \leq 2t + 31$

Show your working clearly.

.....

(Total for Question 7 is 4 marks)

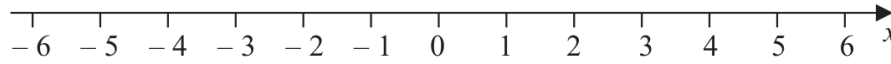
Q22**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 1H • Q9

CH 2 • TOPIC 11 5 marks

- 9 (i) Solve the inequalities $-7 \leq 2x - 3 < 5$

- (ii) On the number line, represent the solution set to part (i)



(Total for Question 9 is 5 marks)

04 Quadratic Inequalities: Sketch and Read

Never solve it like an equation. Find the two roots, sketch the U, then read the answer: > 0 is **outside**, < 0 is **between**.

WHEN YOU SEE → YOU DO

$x^2 \dots > 0$ ("greater than")

→ outside: $x < a$ or $x > b$

$x^2 \dots < 0$ ("less than")

→ between: $a < x < b$

"Show clear algebraic working"

→ show the factorised form and the sketch

terms on both sides

→ ERCS first: make it $\dots > 0$ with $+x^2$

Make it > 0 or < 0

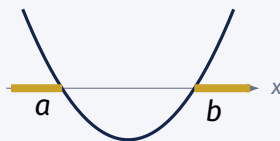
Roots (MODE 5 3)

Sketch the U

Read the region

ZONE A

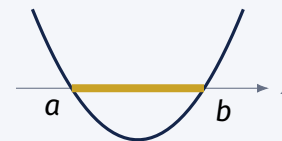
> 0 or ≥ 0 : **OUTSIDE**



$x < a$ or $x > b$ (two answers)

ZONE B

< 0 or ≤ 0 : **BETWEEN**



$a < x < b$ (one answer)

WORKED MODEL • JUNE 2024 P2HR

Solve $6x^2 - 7x - 20 > 0$.

1 $(3x + 4)(2x - 5) = 0 \Rightarrow x = -\frac{4}{3}, \frac{5}{2}$

roots

2 > 0 : the parts of the U above the axis

sketch: outside

ANSWER $x < -\frac{4}{3}$ or $x > \frac{5}{2}$

× **TRAP** $x^2 > 9$ is **not** $x > 3$ only. It is $x < -3$ or $x > 3$ — sketch, do not square-root.

IDEA 04

Practise: Quadratic Inequalities

My questions first, then selected practice, then real past papers.

Q23 DR ESLAM Between

CH 2 • TOPIC 11 3 marks

Solve $x^2 - x - 6 < 0$

Q24 DR ESLAM Outside

CH 2 • TOPIC 11 3 marks

Solve $x^2 - 4x > 0$

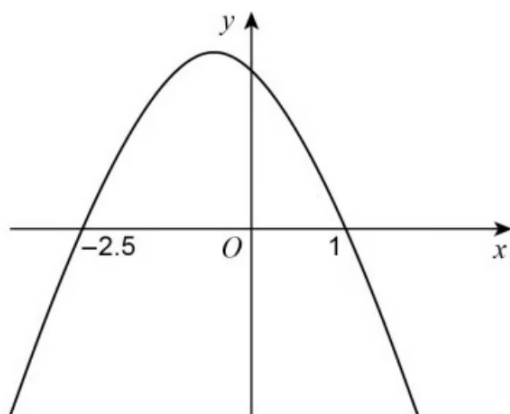
SELECTED PRACTICE

Q25**PRACTICE**

Practice · Easy

CH 2 · TOPIC 11 1 mark

- 14 Here is a sketch of $y = f(x)$ where $f(x)$ is a quadratic function. The graph intersects the x -axis where $x = -2.5$ and $x = 1$



Circle the solution of $f(x) > 0$

$x < -2.5$ or $x > 1$	$x > -2.5$ or $x > 1$
$-2.5 < x < 1$	$x > -2.5$ or $x < 1$

(1 mark)

Q26**PRACTICE**

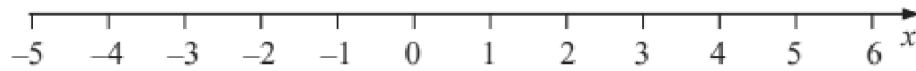
Practice · Medium

CH 2 · TOPIC 11 4 marks

22 (a) Solve $x^2 + 2x > 6x + 5$

(3 marks)

(b) Represent your solution set to part (a) on the number line below.



(1 mark)

Q27**PRACTICE**

Practice · Hard

CH 2 · TOPIC 11 4 marks

4 Solve the inequality $4x^2 - 5x - 6 > 0$

(4 marks)

Q28**PRACTICE**

Practice · Hard

CH 2 · TOPIC 11 3 marks

14 Find the interval for which $x^2 - 7x + 10 \leq 0$.

..... $\leq x \leq$

(3 marks)

PAST PAPERS**Q29****PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 2HR · Q19

CH 2 · TOPIC 11 6 marks

19 (a) Solve $\frac{4-3x}{5} - \frac{3x-5}{2} = -3$

Show clear algebraic working.

$x =$

(b) Solve the inequality $5y^2 - 17y \leq 40$

.....

(Total for Question 19 is 6 marks)

Q30**PAST PAPER**

Edexcel 4MA1 · June 2024 · Paper 2HR · Q20

CH 2 · TOPIC 11**4 marks**

- 20 Solve $6x^2 - 7x - 20 > 0$
Show clear algebraic working.

.....

(Total for Question 20 is 4 marks)

Q31**PAST PAPER**

Edexcel 4MA1 · November 2024 · Paper 2H · Q20

CH 2 · TOPIC 11**3 marks**

- 20 Solve the inequality $10x^2 + 11x - 21 < 0$
Show clear algebraic working.

.....

(Total for Question 20 is 3 marks)

Q32**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 2HR · Q21

CH 2 · TOPIC 11 3 marks

21 Solve the inequality $2x^2 - 7x - 15 > 0$

Show clear algebraic working.

(Total for Question 21 is 3 marks)

Q33**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 2H · Q19

CH 2 · TOPIC 11 3 marks

19 Solve the inequality $4x^2 + 4x - 15 < 0$

Show clear algebraic working.

(Total for Question 19 is 3 marks)

Q34 **PAST PAPER** Edexcel 4MA1 · November 2025 · Paper 1H · Q25**CH 2 · TOPIC 11** **3 marks**

- 25 Solve the inequality $2x^2 + x - 28 > 0$
Show clear algebraic working.

.....

(Total for Question 25 is 3 marks)

05 Problems with Inequalities

Build the inequality from the words, solve it, then use common sense: a length must be positive.

WHEN YOU SEE → YOU DO

“the area is less than 75 cm^2 ”

→ area formula < 75 , then Idea 04

a side $x - 1$

→ also need $x - 1 > 0$, so $x > 1$

“integer such that ... and ...”

→ solve both, list the integers in the overlap

“greatest possible value of $y - x$ ”

→ make y big and x small

WORKED MODEL • JANUARY 2022 P2H

Rectangle $(2x + 3)$ by $(x - 1)$. Area < 75 . Find the range of x .

1 $(2x + 3)(x - 1) < 75$

area < 75

2 $2x^2 + x - 78 < 0$

ERCS

3 $(2x + 13)(x - 6) < 0 \Rightarrow -6.5 < x < 6$

between

4 side $x - 1 > 0 \Rightarrow x > 1$

lengths positive

ANSWER $1 < x < 6$

IDEA 05

Practise: Problems with Inequalities

My questions first, then selected practice, then real past papers.

SELECTED PRACTICE

Q35

PRACTICE

Practice · Hard

CH 2 · TOPIC 11 5 marks

6 n is an integer such that $3n + 2 \leq 14$ and $\frac{6n}{n^2 + 5} > 1$

Find all the possible values of n .

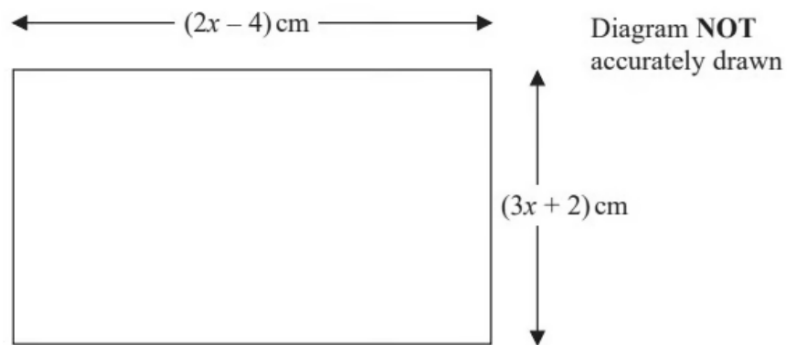
(5 marks)

Q36**PRACTICE**

Practice · Hard

CH 2 · TOPIC 11 5 marks

- 9 The diagram shows a rectangle.



The area of the rectangle is $A \text{ cm}^2$

Given that $A < 3x + 27$

find the range of possible values for x .

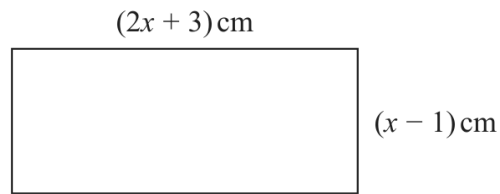
(5 marks)

PAST PAPERS

Q37 PAST PAPER Edexcel 4MA1 • January 2022 • Paper 2H • Q22

CH 2 • TOPIC 11 5 marks

22 Here is a rectangle.

Diagram **NOT**
accurately drawnGiven that the area of the rectangle is less than 75 cm^2 find the range of possible values of x

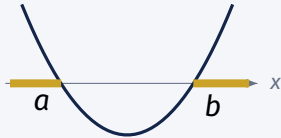
(Total for Question 22 is 5 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

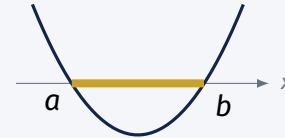
ZONE A

> 0 : **OUTSIDE**



ZONE B

< 0 : **BETWEEN**



01 Number lines

$<$, $>$ open circle; $\leq$, $\geq$ filled. List every integer, check both ends.

02 Linear

Like an equation; $\times$ or $\div$ by a negative flips the sign.

03 Double

Do it to all three parts. Two inequalities: keep the overlap.

04 Quadratic

Roots $\rightarrow$ sketch the U $\rightarrow > 0$ outside, < 0 between.

05 Problems

Build it, solve it, lengths > 0 .

BOTTOM LINE Solve like an equation $\rightarrow$ negative? flip $\rightarrow$ quadratic? roots, sketch, read.

Answers

Solving Inequalities • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • NUMBER LINES AND INTEGERS

- Q1** Dr Eslam • List the integers $-2 \leq n < 3$: $-2, -1, 0, 1, 2$
- Q2** Practice • Easy (a) $-1, 0, 1, 2, 3$ (b) $-4 < x \leq 3$ (c) $y > 7$
- Q3** Practice • Easy $-4 < x \leq 5$
- Q4** Practice • Medium (a) open at -2 , filled at 3 (b) $x \geq 3.5$
- Q5** Practice • Medium (a) $x \geq 4.5$ (b) $2 \leq x < 5$
- Q6** Edexcel 4MA1 • January 2021 • Paper 1HR • Q3 (a) filled at -2 , open at 1 (b) $-3, -2, -1, 0, 1, 2$
- Q7** Edexcel 4MA1 • January 2022 • Paper 1HR • Q1 (a) $-2, -1, 0, 1, 2$ (b) filled at 1 , arrow left
- Q8** Edexcel 4MA1 • June 2025 • Paper 1HR • Q6 (a) $-2 < x \leq 1$ (b) $a \leq 8.25$

IDEA 02 • SOLVE A LINEAR INEQUALITY

- Q9** Dr Eslam • Keep the sign $x < 5$
- Q10** Dr Eslam • Flip the sign $-3x \geq 9; x \leq -3$
- Q11** Practice • Easy (a) $n > 2$ (b) $-5 < x \leq 1$: open at -5 , filled at 1
- Q12** Practice • Easy $x > 4$
- Q13** Practice • Medium $x < 5$: open at 5 , arrow left
- Q14** Practice • Hard No: $x \leq -3$, the arrow should point left
- Q15** Edexcel 4MA1 • January 2021 • Paper 2H • Q5 (a) $2x^2 + 29x$ (b) $n = 14$ (c)(i) $t < 3$ (ii) open at 3 , arrow left
- Q16** Edexcel 4MA1 • June 2022 • Paper 1HR • Q9 $x \geq -2$

IDEA 03 • DOUBLE INEQUALITIES

- Q17** Dr Eslam • All three parts $-6 < 3x \leq 9; -2 < x \leq 3$
- Q18** Practice • Medium (a) $x > 2.6$ (b) $-2, -1, 0, 1, 2, 3$
- Q19** Practice • Medium $-1, 0, 1, 2$
- Q20** Practice • Hard $x = -2, y = 13: 15$
- Q21** Edexcel 4MA1 • May 2021 • Paper 1H • Q7 (a) $-2, -1, 0, 1, 2$ (b) $t \leq 6.8$
- Q22** Edexcel 4MA1 • November 2021 • Paper 1H • Q9 (i) $-2 \leq x < 4$ (ii) filled at -2 , open at 4

IDEA 04 • QUADRATIC INEQUALITIES

- Q23** Dr Eslam • Between $(x - 3)(x + 2) < 0; -2 < x < 3$
- Q24** Dr Eslam • Outside $x(x - 4) > 0; x < 0 \text{ or } x > 4$
- Q25** Practice • Easy $-2.5 < x < 1$
- Q26** Practice • Medium (a) $x < -1 \text{ or } x > 5$
- Q27** Practice • Hard $x < -\frac{3}{4} \text{ or } x > 2$
- Q28** Practice • Hard $2 \leq x \leq 5$
- Q29** Edexcel 4MA1 • January 2020 • Paper 2HR • Q19 (a) $x = 3$ (b) $-1.6 \leq y \leq 5$
- Q30** Edexcel 4MA1 • June 2024 • Paper 2HR • Q20 $x < -\frac{4}{3} \text{ or } x > \frac{5}{2}$
- Q31** Edexcel 4MA1 • November 2024 • Paper 2H • Q20 $(10x + 21)(x - 1) < 0; -2.1 < x < 1$
- Q32** Edexcel 4MA1 • June 2025 • Paper 2HR • Q21 $x < -1.5 \text{ or } x > 5$
- Q33** Edexcel 4MA1 • June 2025 • Paper 2H • Q19 $-2.5 < x < 1.5$
- Q34** Edexcel 4MA1 • November 2025 • Paper 1H • Q25 $x < -4 \text{ or } x > 3.5$

IDEA 05 • PROBLEMS WITH INEQUALITIES

- Q35** Practice • Hard $n \leq 4 \text{ and } 1 < n < 5: 2, 3, 4$
- Q36** Practice • Hard $6x^2 - 11x - 35 < 0, -\frac{5}{3} < x < \frac{7}{2}; \text{ with } x > 2:$
 $2 < x < 3.5$
 $1 < x < 6$
- Q37** Edexcel 4MA1 • January 2022 • Paper 2H • Q22